Europe's Uneven Progress on $PM_{2.5}$: Interactive Data Story

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Abstract

This report describes the development of an interactive data story about changes in fine particulate matter ($PM_{2.5}$) across Europe between 2007 and 2024. The project uses annual interpolated raster datasets published by the European Environment Agency. Following the visual data storytelling process proposed by Lee et al., the work progressed from data exploration and preprocessing to narrative design, implementation, and evaluation. A Martini Glass structure guides readers through the main findings before allowing independent exploration. Narrative patterns and transitions communicate spatial and temporal changes, while the final design combines guided storytelling, interactive maps, and methodological transparency.

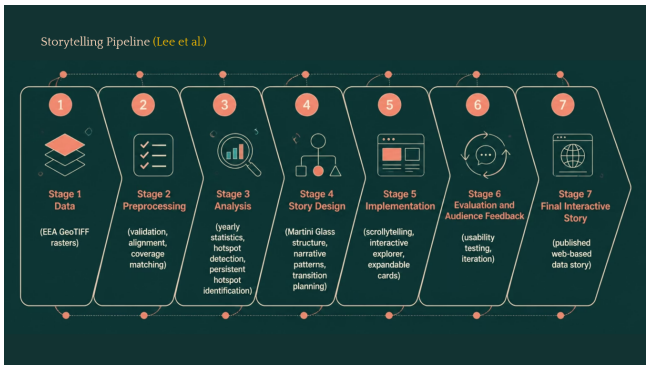


Figure 1: Storytelling Pipeline used in this project.

1 Introduction

Fine particulate matter, or $PM_{2.5}$, is particularly relevant to public health because the particles are small enough to penetrate deep into the lungs and can enter the bloodstream [3]. Although air quality has improved in many parts of Europe, this progress has not been spatially uniform. A single European average can therefore hide persistent regional differences.

The goal of this project was to communicate these long-term spatial and temporal patterns through an interactive data story. Instead of a dashboard, the project guides readers through key findings before opening an interactive explorer. The intended audience is the general public with limited experience interpreting environmental maps.

The development was organised around the visual data storytelling process described by Lee et al., which connects the exploration of data, the construction of a narrative, and the communication of that narrative to an audience [1]. Figure 1 summarises how this process was applied in the project.

2 Development Process

2.1 Exploring and Preparing the Data

The project uses annual $PM_{2.5}$ concentration rasters published by the European Environment Agency for 2007–2024, with 2009 unavailable in the selected series. The datasets describe estimated annual average concentrations on a one-square-kilometre grid generated from monitoring observations, atmospheric modelling, and spatial interpolation [2].

Raster data was selected instead of station-only data because it provides continuous spatial coverage, making regional comparisons across Europe easier. The story also explains that raster cells represent modelled estimates rather than direct measurements.

The preprocessing stage included inspecting raster properties, transforming yearly files for browser visualization, and generating summary statistics to analyse long-term spatial changes.

Early exploration showed that the story was not simply about overall improvement, but about uneven progress across different regions. This became the project's main narrative.

2.2 From Analysis to Story

A Martini Glass narrative structure was chosen because it combines guided storytelling with reader exploration. The opening sections introduce $PM_{2.5}$, explain the data, and present key comparisons before handing control to an interactive explorer.

A dashboard was considered but would have required readers to interpret the data with little context. The chosen structure first establishes the main argument before encouraging independent exploration.

The story follows a setup, rising action, climax, and resolution, moving from a European overview to regional differences before concluding with the interactive explorer.

2.3 Narrative Patterns and Transitions

Several narrative patterns were used to support the story. *Familiarization* explains the pollutant, its health relevance, and the transformation from station observations to raster estimates before asking readers to interpret maps. *Gradual Reveal* limits each section to one main idea so that methodology, trend, and regional variation do not compete for attention.

Comparison supports the central argument by placing different years and regions in a shared visual context. *Concretize* connects Europe-wide patterns to selected regional examples, preventing the story from remaining too abstract. *Emphasis* directs attention to important locations and values through annotations and highlighted states. A short *Recap* before the explorer reminds readers of the main findings and explains what they can investigate independently.

The story also uses the transition terminology introduced in the course. *Comparison transitions* connect yearly states and regional

values. *Spatial transitions* move attention between areas of Europe. *Granularity transitions* move from the continental overview toward regional and local detail. These transitions were used deliberately so that changes in viewpoint reflected the argument rather than functioning as decoration.

2.4 Implementation Decisions

The final prototype was implemented as a browser-based scrollytelling experience. Explanatory text is connected to map states that change as the reader progresses. The guided portion uses a stable visual layout to make comparisons easier and to reduce unnecessary interface changes.

The final explorer adds controls for year and location-based investigation. Displaying the complete raster directly would have introduced performance and interaction problems because each annual dataset contains a very large number of cells. The browser representation was therefore simplified for responsive interaction while preserving the main spatial distribution. This is stated as a technical limitation rather than presented as equivalent to the complete analytical raster.

Expandable methodology cards were added to keep the main story concise. Readers can continue through the narrative without interruption, while those interested in the source, interpolation process, or analysis can inspect further details. This decision balances readability with transparency.

A future-oriented view was included as an exploratory trend-continuation scenario. It should not be interpreted as a validated forecast. Its purpose is to ask what the spatial pattern might look like if historical improvement continued, not to predict future concentrations with certainty.

2.5 Evaluation and Iteration

The prototype was revised through tutorial feedback, intermediate presentations, and informal audience testing. Earlier versions placed too much emphasis on a broad claim that Europe had improved. Feedback showed that this was already familiar and did not provide a sufficiently distinctive story. The project was therefore refocused on uneven progress, persistent differences, and the relationship between the continental pattern and regional detail.

Other revisions reduced introductory text, clarified the meaning of raster cells, and improved the transition from guided sections to the explorer. Methodological explanations were separated from the primary narrative through expandable cards. Labels and annotations were also revised to distinguish measured station observations, interpolated annual estimates, and derived scenarios.

These changes improved both narrative clarity and trust. In particular, explicitly describing the data as interpolated estimates prevents the maps from suggesting a level of measurement precision that the underlying data does not provide.

3 Discussion

The strongest reason for the final design is that the topic requires both explanation and exploration. A fully linear story could communicate the main argument clearly but would limit readers who want to investigate their own region. A dashboard could support exploration but would offer too little guidance. The Martini Glass structure provides a useful middle ground by establishing shared context before interaction begins.

The maps were not included only because the data is geographical. They directly support the argument that European progress is spatially uneven. Temporal comparisons show whether concentrations changed, while spatial and granularity transitions show where those changes occurred and whether a continental summary holds at regional level.

The project has several limitations. The maps use annual interpolated estimates and do not represent direct measurements at every raster cell. The interactive representation simplifies the full-resolution data for browser performance. The future-oriented visualization is a scenario based on continuation of historical change and is not a formal forecasting model. Finally, annual means are useful for long-term comparison but do not show short pollution episodes or seasonal variation.

Despite these limitations, the final design supports the central communication goal. It gives readers enough methodological context to interpret the maps, guides them toward a clear argument, and then allows them to test that argument through their own exploration.

4 Conclusion

This project applied the visual data storytelling process to a large environmental raster dataset. Data exploration first identified uneven progress as the central story. The Martini Glass structure then combined an author-guided narrative with an interactive explorer. Familiarization, comparison, gradual reveal, emphasis, and recap were used alongside comparison, spatial, and granularity transitions to structure the reader's experience.

The final result does not attempt to replace detailed environmental analysis. Instead, it translates a complex multi-year dataset into an accessible story while documenting important analytical and technical limitations. The project shows how narrative structure and interaction can work together to communicate spatial change without removing the reader's ability to explore the data independently.

References

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